Spring 4 SPM

Targets

Target 1: AI Analysis Latency & Inference Degradation

- **Indicator (SLI Type):** Latency
- **Technical Definition:** The end-to-end P95 latency measured from the client payload transmission of an unstructured expense string or financial query to the API gateway, through downstream Large Language Model (LLM) token inference/categorization, down to successful DOM rendering of structured budget insights on the client dashboard.
- **Measurement Methodology:**
    - **Tool:** Datadog / APM distributed tracing coupled with automated browser testing scripts checking real-time response round-trips.
    - **Schedule:** Continuously monitored in staging/production, aggregated and reviewed every Monday and Friday.
    - **Owner:** Frontend / AI Integration Lead.
- **SLO (Target & Time Window):** 95% of asynchronous AI-driven categorization and insight generation requests must complete and render within a 3.0-second window over a rolling 30-day sprint cycle.
- **Master-Level Rule & Error Budget Justification:**
    - *Embracing Risk:* This yields a 5% error budget for slow responses per sprint.
    - *Rule Check:* Adheres to "perfection can wait" (95% threshold prevents brittle, over-engineered optimizations); avoids unmeasurable absolute phrasing.
- **Who Would Notice:** End-users dependent on responsive financial tracking, where high latency degrades user retention and increases transactional friction.

Target 2: Financial Arithmetic Correctness & Deterministic Transaction Sync

- **Indicator (SLI Type):** Correctness

- **Technical Definition:** The ratio of mathematically deterministic financial calculations (budget totals, delta allocations, tax forecasts) and ledger API sync actions that execute without unhandled floating-point exceptions, race conditions, or LLM-induced hallucinated data mutations against user accounts.
- **Measurement Methodology:**
    - **Tool:** Automated integration test suites executing transactional mock payloads via GitHub Actions CI/CD pipelines, alongside post-deployment invariant assertions.
    - **Schedule:** Executed automatically on every pull request merge and enforced strictly prior to every sprint deployment deadline.
    - **Owner:** Backend / QA Lead.
- **SLO (Target & Time Window):** 100% calculation accuracy with zero unhandled arithmetic errors or broken ledger synchronization states at every sprint deployment deadline.
- **Master-Level Rule & Error Budget Justification:**
    - *Zero-Tolerance Policy:* Adopts a strict zero-tolerance target (0% allowed budget), meaning any single financial discrepancy or silent failure instantly exhausts the error budget, triggering an automatic rollback and mandatory regression review.
    - *Rule Check:* Chosen deliberately for financial domains where data integrity is non-negotiable.
- **Who Would Notice:** App users and the instructor acting as project sponsor, who require absolute ledger accuracy to prevent monetary miscalculations.